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Section: 04

1. Network request screenshots from Chrome DevTools showing successful API calls and response JSON:

Successful API calls:

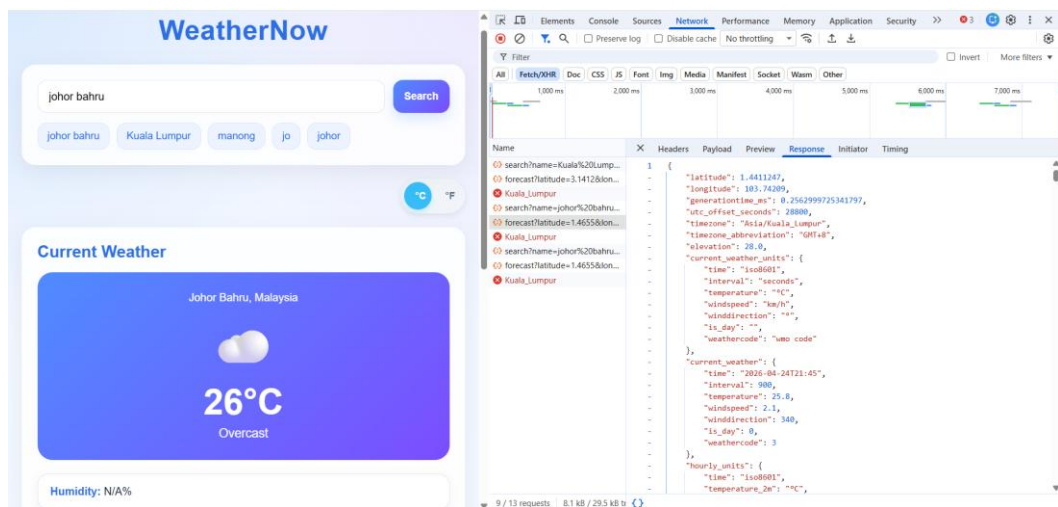
The screenshot displays the WeatherNow application interface on the left and the Chrome DevTools Network tab on the right. The app shows a search bar with 'johor bahu' and a 'Search' button. Below the search bar, there are buttons for 'johor bahu', 'Kuala Lumpur', 'manong', 'jo', and 'johor'. The 'Current Weather' section shows 'Johor Bahru, Malaysia' with a temperature of 26°C and 'Overcast' conditions. The humidity is listed as 'N/A%'. The Network tab shows a list of requests, including 'search?name=Kuala%20Lumpu&count=1&language=...' and 'forecast?latitude=3.1412&longitude=101.68653&lon...'. The status of these requests is '200', indicating successful API calls.

Call Open-Meteo Geocoding:

This screenshot shows the WeatherNow app interface on the left and the Chrome DevTools Network tab on the right, specifically focusing on the 'Response' tab for the 'search?name=Kuala%20Lumpu...' request. The app interface is identical to the previous screenshot. The Network tab shows the response JSON for the geocoding API call, which includes details about Johor Bahru, Malaysia, such as its ID, name, latitude, longitude, elevation, and administrative divisions. The response is structured as follows:

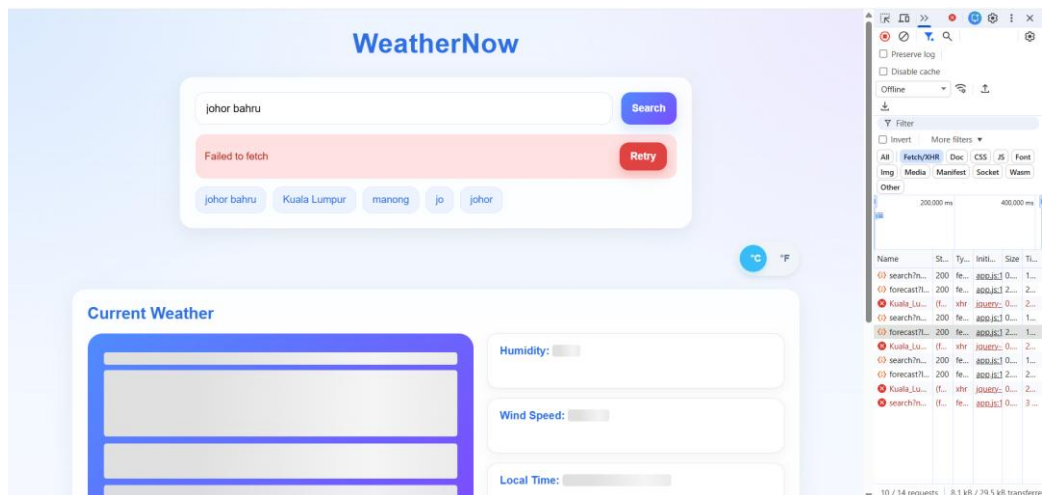
```
{
  "results": [
    {
      "id": 1732752,
      "name": "Johor Bahru",
      "latitude": 1.4655,
      "longitude": 103.7578,
      "elevation": 32.0,
      "feature_code": "PPLA",
      "country_code": "MY",
      "admin1_id": 1733849,
      "admin2_id": 1732751,
      "timezone": "Asia/Kuala_Lumpur",
      "population": 858118,
      "country_id": 1733045,
      "country": "Malaysia",
      "admin1": "Johor",
      "admin2": "Daerah Johor Bahru"
    }
  ],
  "generationtime_ms": 0.5661249
}
```

Call Open-Meteo:

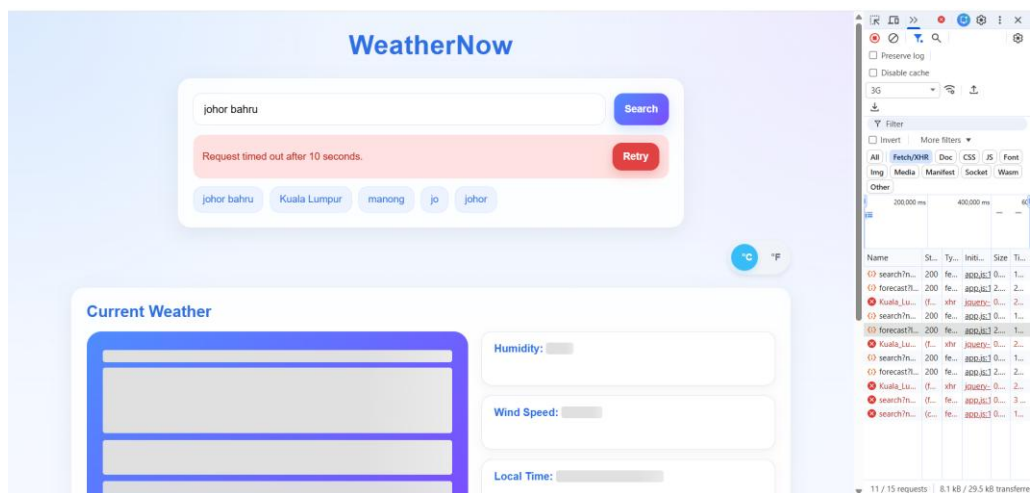


2. Screenshot of the error state UI (network offline simulation via DevTools)

Offline Mode:



Slow Internet Mode:



3. Reflection comparing Fetch API vs jQuery AJAX

Both Fetch API and jQuery AJAX are used to make HTTP requests. In this Lab 3, the Fetch API was used to fetch the geocoding and weather API calls, while jQuery's `$.getJSON()` was used to fetch the local time from the WorldTimeAPI.

The Fetch API is a built-in JavaScript feature that provides a modern, promise-based approach to handling requests. It uses the *async/await* which promote cleaner and more readable code. In error handling, it requires developers to manually check response status codes. In addition, It requires more manual steps for time configuration as the responses have to parse explicitly.

jQuery AJAX is part of the jQuery library and relies on callback functions. It uses methods like *.done()*, *.fail()*, and *.always()*, which clearly separate success, failure, and completion logic. In term of error handling, jQuery AJAX automatically handles HTTP errors through its built-in callbacks. Besides, it supports automatic JSON parsing and built-in timeout configuration.